

§1.4

P1. $\frac{1}{1+y^2} \frac{dy}{dt} = \frac{1}{1+t^2}$

$$F(y) = \int \frac{1}{1+y^2} dy = \arctan y$$

so we have.

$$\arctan y = \int \frac{1}{1+t^2} dt + C$$

$$= \arctan t + C$$

$$\Rightarrow y(t) = \tan(\arctan t + C)$$

$$= \frac{t + \tan C}{1 - t \cdot \tan C}$$

P2. $\frac{1}{1+y} \frac{dy}{dt} = 1+t$

$$F(y) = \int \frac{1}{1+y} dy = \ln|1+y|$$

so we have.

$$\ln|1+y| = \int 1+t dt$$

$$= t + \frac{t^2}{2} + C_1$$

$$|1+y| = e^{t + \frac{t^2}{2} + C_1}$$

$$\Rightarrow \underbrace{|(1+y) e^{-t - \frac{t^2}{2}}|}_{\text{continuous}} = \underbrace{e^{C_1}}_{\text{Constant}} = C_2$$

By the same argument in Page 4, we

have. for some C

$$(1+y) e^{-t - \frac{t^2}{2}} = C$$

$$1+y = C e^{t + \frac{t^2}{2}}$$

$$y(t) = C e^{t + \frac{t^2}{2}} - 1$$

P4. $\frac{dy}{dt} = e^{t+3} y$

$$e^{-y} \frac{dy}{dt} = e^{t+3}$$

$$F(y) = \int e^{-y} dy = -e^{-y}$$

So we have

$$-e^{-y} = \int e^{t+3} dt + C$$

$$= e^{t+3} + C$$

$$e^{-y} = -e^{t+3} - C$$

$$y(t) = -\ln(-e^{t+3} - C)$$

P5. $\frac{\cos y}{\sin y} \frac{dy}{dt} = \frac{\cos t}{\sin t}$

$$F(y) = \int \frac{\cos y}{\sin y} dy = \ln|\sin y|$$

so we have.

$$\ln|\sin y| = \int \frac{\cos t}{\sin t} dt + C_1$$

$$= \ln|\sin t| + C_1$$

$$\ln \left| \frac{\sin y}{\sin t} \right| = C_1$$

$$\left| \frac{\sin y}{\sin t} \right| = e^{C_1} = C_2$$

Again, we obtain by the same argument

$$\frac{\sin y}{\sin t} = C \quad \text{for some } C \text{ in Page 4.}$$

$$\sin y = C \sin t$$

$$y(t) = \arcsin(C \sin t)$$

P6. $\frac{2y}{1+y^2} \frac{dy}{dt} = -t^2, \quad y(0) = 1$

$$\int_{y(0)}^{y(t)} \frac{2r}{1+r^2} dr = \int_0^t -s^2 ds$$

$$\ln(1+r^2) \Big|_{r=1}^{r=y(t)} = -\frac{s^3}{3} \Big|_{s=0}^{s=t}$$

$$\ln(1+y^2(t)) - \ln 2 = -\frac{t^3}{3}$$

$$\frac{1+y^2(t)}{2} = e^{-\frac{t^3}{3}}$$

$$y(t) = \pm \sqrt{2e^{-\frac{t^3}{3}} - 1}$$

Notice that $y(0) = 1 > 0$, so we know

$$y(t) = \sqrt{2e^{-\frac{t^3}{3}} - 1}$$

This $y(t)$ is only defined when

$$2e^{-\frac{t^3}{3}} - 1 \geq 0$$

$$e^{-\frac{t^3}{3}} \geq \frac{1}{2}$$

$$e^{\frac{t^3}{3}} \leq 2$$

$$\frac{t^3}{3} \leq \ln 2$$

$$t \leq (3 \ln 2)^{\frac{1}{3}}$$

Therefore, $y(t)$ only exists on the interval $(-\infty, (3 \ln 2)^{\frac{1}{3}}]$.

P7. $y \frac{dy}{dt} = \frac{2t}{1+t^2}$, $y(2) = 3$

$$\int_{y(2)}^{y(t)} r dr = \int_2^t \frac{2s}{1+s^2} dt$$

$$\frac{y^2(t)}{2} - \frac{3^2}{2} = \ln(1+s^2) \Big|_{s=2}^{s=t}$$

$$\frac{y^2(t)}{2} - \frac{9}{2} = \ln \frac{1+t^2}{5}$$

$$y^2(t) = 2 \ln \frac{1+t^2}{5} + 9$$

$$y(t) = \pm \sqrt{2 \ln \frac{1+t^2}{5} + 9}$$

Notice that $y(2) = 3 > 0$, so we have.

$$y(t) = \sqrt{2 \ln \frac{1+t^2}{5} + 9}$$

and $y(t)$ exists on the whole $(-\infty, +\infty)$

P9. $2(y-1) \frac{dy}{dt} = 3t^2 + 4t + 2$, $y(0) = -1$

$$\int_{y(0)}^{y(t)} 2(r-1) dr = \int_0^t (3s^2 + 4s + 2) ds$$

$$(r-1)^2 \Big|_{r=-1}^{r=y(t)} = (s^3 + 2s^2 + 2s) \Big|_{s=0}^{s=t}$$

$$(y(t)-1)^2 - 4 = t^3 + 2t^2 + 2t$$

$$(y(t)-1)^2 = t^3 + 2t^2 + 2t + 4$$

$$y(t) = \pm \sqrt{t^3 + 2t^2 + 2t + 4} + 1$$

Notice that $y(0) = -1$, so we know that

$$y(t) = -\sqrt{t^3 + 2t^2 + 2t + 4} + 1$$

This $y(t)$ is only defined when.

$$t^3 + 2t^2 + 2t + 4 \geq 0$$

$$(t+2)(t^2+2) \geq 0$$

$$t \geq -2$$

Therefore, $y(t)$ only exists on the interval $[-2, +\infty)$

P10. $\frac{\cos y}{\sin y} \frac{dy}{dt} = \frac{-t}{1+t^2}$, $y(1) = \frac{\pi}{2}$

$$\int_{y(1)}^{y(t)} \frac{\cos r}{\sin r} dr = \int_1^t \frac{-s}{1+s^2} ds$$

$$\ln|\sin r| \Big|_{r=\frac{\pi}{2}}^{r=y(t)} = -\frac{1}{2} \ln(1+s^2) \Big|_{s=1}^{s=t}$$

$$\ln|\sin y(t)| = \frac{1}{2} \ln \frac{2}{1+t^2} \Rightarrow y(t) = \arcsin\left(\pm \sqrt{\frac{2}{1+t^2}}\right)$$

Notice that $y(1) = \frac{\pi}{2}$, so we know

$$y(t) = \arcsin \sqrt{\frac{2}{1+t^2}}$$

Domain of t : $\sqrt{\frac{2}{1+t^2}} \leq 1 \Rightarrow \frac{2}{1+t^2} \leq 1 \Rightarrow t \geq 1$.

P14. (a) $\frac{y^2 + 2ty}{y^2} = 1 + \frac{2t}{y} = 1 + \frac{2}{(\frac{y}{t})}$, Yes.

(b) $\frac{y^3 + t^3}{yt^2 + y^3} = \frac{(\frac{y}{t})^3 + 1}{(\frac{y}{t}) + (\frac{y}{t})^3}$, Yes.

(c) $\ln y - \ln t + \frac{t+y}{t-y}$
 $= \ln \frac{y}{t} + \frac{1 + (\frac{y}{t})}{1 - (\frac{y}{t})}$, Yes.

P16. $2ty \frac{dy}{dt} = 3y^2 - t^2.$

$$2\left(\frac{y}{t}\right) \frac{dy}{dt} = 3\left(\frac{y}{t}\right)^2 - 1 \quad \text{--- ①}$$

Let $u(t) = \frac{y(t)}{t}$, i.e., $y(t) = tu(t)$, then

$$\frac{dy}{dt} = u(t) + t \frac{du}{dt}$$

so ① can be transformed into

$$2u\left(u + t \frac{du}{dt}\right) = 3u^2 - 1$$

$$t \frac{du}{dt} = \frac{3u^2 - 1}{2u} - u$$

$$t \frac{du}{dt} = \frac{u^2 - 1}{2u}$$

$$\frac{2u}{u^2 - 1} \frac{du}{dt} = \frac{1}{t}$$

$$F(u) = \int \frac{2u}{u^2 - 1} du = \ln|u^2 - 1|$$

So we have

$$\ln|u^2 - 1| = \int \frac{1}{t} dt + C_1$$

$$\ln|u^2 - 1| = \ln|t| + C_1$$

$$\ln\left|\frac{u^2 - 1}{t}\right| = C_1 \Rightarrow \left|\frac{u^2 - 1}{t}\right| = e^{C_1} = C_2$$

By the same argument in Page 4, we have.

$$\frac{u^2 - 1}{t} = C, \text{ for some } C.$$

Replace u by $\frac{y}{t}$, we get

$$\frac{\frac{y^2}{t^2} - 1}{t} = C$$

$$\frac{y^2 - t^2}{t^3} = C.$$

that is the solution curve of this problem.